CS258

THE UNIVERSITY OF WARWICK

Second Year Winter Examinations

2018/19 Database Systems

Time allowed: 2 hours

Answer **FOUR** questions out of six.

Read carefully the instructions on the answer book and make sure that the particulars required are entered on **each** answer book.

Calculators are not required and not permitted.

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1. Consider the COMPANY schema diagram in Figure 1 overleaf.

   a. Explain the meaning of the following phrase.

      Pno *is a foreign key of* WORKS_ON *that references* PROJECT.

      [2]

   b. Write SQL code to create a view named V that provides the following details for every department: department number, manager's last name, average departmental salary.

      [5]

   c. Is it possible to define a view that will report the following details for each employee: Ssn and the number of employees who work on exactly the same set of projects? Explain why not or provide an SQL query.

      [10]

   d. Write SQL code to create a view V that has the project name, controlling department name, number of employees, and total hours worked per week on the project for each project with more than one employee working on it.

      [8]

   Total mark for this question: 25

Continued

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date

DEPT_LOCATIONS

Dnumber	Dlocation

PROJECT

Pname	Pnumber	Plocation	Dnum

WORKS_ON

Essn	Pno	Hours

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship

Figure 1: COMPANY schema diagram

Continued

2. Write the following queries in SQL for the COMPANY database of Figure 1.

   a. Retrieve the birth date of the manager of Department No. 6.

   [3]

   b. Retrieve the first and last names of all employees whose salary is greater than that of the manager of Department No. 5.

   [5]

   c. Retrieve all distinct project numbers such that everybody working on them has 'A.' as the middle initial.

   [7]

   d. For all departments with a male employee but without any female employees, list the department number followed by the number of male employees.

   [10]

   Total mark for this question: 25

3. Specify the following queries in relational algebra for the COMPANY database (Figure 1).

   a. Retrieve Social Security Numbers of all employees whose first name is 'Graham'.

   [2]

   b. Retrieve names of Alan J. Baker's dependents.

   [5]

   c. Retrieve the last names of all employees, which are supervised by the manager of Department No. 3.

   [8]

   d. Find the last names of employees who work on all the projects controlled by Department No. 1.

   [10]

   Total mark for this question: 25

Continued

4.

a. Give the definition of a key.

[2]

b. Consider the relation schema PERSON(*age, first name, surname, postcode*) and the relation state given below. Identify potential keys of PERSON.

[6]

age	*first name*	*surname*	*postcode*
40	Barbara	Taylor	CV4 7AL
40	Bernard	Jones	CV3 8AK
41	Brenda	Taylor	CV4 7AM
42	Boris	Smith	CV5 2XY
43	Barbara	Taylor	CV4 7XX
44	Boris	Smith	CV1 5GH

c. Which of the following dependencies may hold in the above relation? If a dependency cannot hold, specify the tuples that cause the violation.

i. *age* → *first name*

[2]

ii. *first name* → *surname*

[2]

iii. *surname* → *first name*

[2]

d. Is it true that $X \rightarrow YZ$ imply $X \rightarrow Z$? Justify your answer.

[11]

Total mark for this question: 25

5. Consider the following relation for published books:

BOOK (Book_title, Author_name, Book_type, Listprice, Author_affil, Publisher), where Author_affil refers to the affiliation of the author. Suppose the following dependencies exist:

Book_title -> Publisher, Book_type

Book_type -> Listprice

Author_name -> Author_affil

a. What normal form is the relation in? Explain your answer.

[10]

b. Apply normalization until you cannot decompose the relations further. State the reasons behind each decomposition.

[15]

Total mark for this question: 25

Continued

6. Provide and briefly explain your answer in the following questions.

   a. When (tuples) records have fields (attributes) of variable size, is it true that blocks will have a varying block factor?

   [3]

   b. Consider a DBMS using a heap file organisation, where the blocking factor is F and the file having b blocks.

      i. How many disk blocks are in general required to search for a given record i?

      [3]

      ii. How many blocks are required when contiguous allocation with fixed-length unspanned records is used?

      [3]

   c. Consider an ordered data file with a primary index on its primary key field and a secondary index on another key field. Which of the two indexes brings about the biggest savings in accessing times? Explain with 1-2 sentences.

   [6]

   d. Consider an ordered data file with r = 1,000,000 records; each record has a size of R = 100 bytes. The file is stored on a disk with a block size B = 4,000 bytes. Suppose a primary index has been constructed for the file, using key values of size V = 4 bytes and block pointers are 4 bytes long). How many disk block accesses would be saved by using the index?

   [10]

Total mark for this question: 25

End

6. Provide and briefly explain your answer in the following questions:

   a. When (tuples) records have fields (attributes) of variable size, is it true that blocks will have a varying block factor?

   [2]

   b. Consider a DBMS using a heap file organisation, where the blocking factor is 7 and the file having 8 blocks.

      i. How many disk blocks are in general required to search for a given record?

      [2]

      ii. How many blocks are required when contiguous allocation with fixed-length unspanned records is used?

      [2]

   c. Consider an ordered data file with a primary index on its primary key field and a secondary index on another key field. Which of the two indexes brings about the biggest savings in accessing times? Explain in 1-2 sentences.

   [6]

   d. Consider an ordered data file with $r = 1{,}000{,}000$ records; each record has a size of $R = 150$ bytes. The file is stored on a disk with a block size $B = 4{,}000$ bytes. Suppose a primary index has been constructed for the file, using key values of size $V = 4$ bytes and block pointers are 4 bytes long). How many disk block accesses would be saved by using the index?

   [10]

Total mark for this question: 25

End